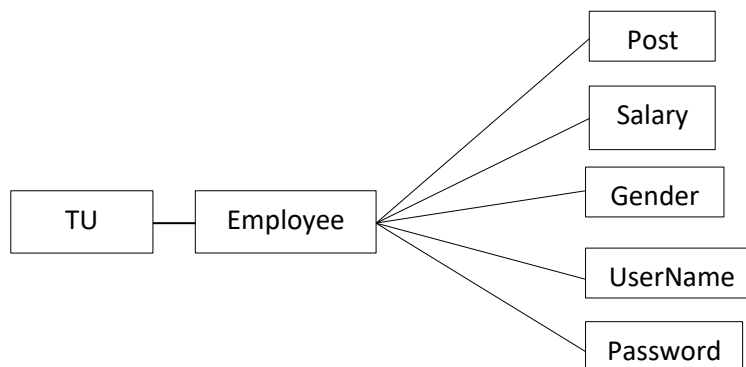


Unit-3: The Client Tier

1. What is XML? Differentiate XML schema with DTD.
2. What are the rules for creating XML document? Write a XML code to store following information about student.
3. What are the rules for well-formed xml document? Create a simple well-formed XML and validate it with DTD Schema.
4. Write XML schema for library information system.
5. What is the use of XML namespaces? Explain with suitable example.
6. Explain the various types of XML schema data types used.
7. What is well formed XML? Explain different ways of DTD implementation for XML documents.
8. Define XPath? Explain different elements required to create XSLT document.
9. How XSLT is differ from CSS? Explain.
10. What are the rules for creating XML document? Write XML code to store following information about student.
 - ❖ Each student has a name, address, phone and website element.
 - ❖ Address must appear multiple times.
 - ❖ Address has attribute named "Type" with value permanent and temporary must be 10 digit.
11. Explain the key component of XML file. Write an XML to describe "address" as an element and "state, city, municipality and house-no" as it's attributes. Also define XSD for the XML file.
12. Make well-formed XML document as per the diagram below, then use an appropriate XSD document to validate it in consideration with below restrictions.



XSD Restrictions:

- a) The username must have a minimum of 8 characters.
- b) The character limits for posts is 5 to 8.
- c) The password must be 8 characters long and begin with Aa – Zz.
- d) The gender must be one of male, female and other.
- e) The salary should range from Rs. 25,000 to Rs. 50,000.

13. Give an XML document that contains a list of products with the following elements: ***productname***, ***category***, ***price*** and ***quantity***, write an XSLT style sheet that:

- a) Filters out any products that have a quantity less than 10.
 - b) Groups the remaining product by category and create a new XML document with a new element called "category" that contains the product within that category as child elements.
 - c) Within each category element, sort the product by price in descending order.
 - d) Transform the element "productname" into an attribute of each product element.
 - e) Add new element called "total_price" to each product element, which contains the total price of the product ($\text{price} \times \text{quantity}$).
14. Write a validate and well-formed xml to store details of visitor list of ABC college with the following details and validate record with the following information.
- a) Record must include details of name, mobile, department, related contact person, data, entry time, exit time, and visitor number as attribute.
 - b) Mobile must be 10 digits and visitor may not have any contact person.
 - c) Record must belong to Admin, IT, Account and Exam Department.
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